

Md Nur Amin Sifat

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🌐 [linkedin.com/in/nur-amin-sifat](https://www.linkedin.com/in/nur-amin-sifat) | 🐙 github.com/GenesisBlock3301 | 📄 leetcode.com/GenesisBlock3301

📄 kaggle.com/genesisblock3301 | 📄 <https://www.researchgate.net/profile/Md-Nur-Amin-Sifat>

📍 Dhaka, Bangladesh

Research Interests

Machine Learning · Deep Learning · NLP · Computer Vision · LLMs · RAG · Generative AI

Research Experience

AI & Computer Vision Intern, Intelsense AI – Dhaka, Bangladesh Jun 2020 – Oct 2020

- Worked on computer vision tasks involving image processing, object detection, and human identification using deep learning techniques.
- Applied Convolutional Neural Networks (CNNs) for improving accuracy in human recognition and gait analysis.
- Gained hands-on experience in real-world applications of AI and contributed to research-driven development workflows.

Work Experience

Software Engineer April 2022 – Present

Brain Station 23, Dhaka, Bangladesh

- Designed and developed microservices for the Waadaa e-commerce platform, focusing on scalability, maintainability, and modular architecture.
- Optimized backend operations, reducing execution time from 700 milliseconds to 2 milliseconds, significantly enhancing system performance.
- Collaborated with cross-functional teams to deliver robust backend solutions in an agile environment.
- Technologies used: **Python**, JavaScript;
- Frameworks: **Django**, **Django REST Framework**, **FastAPI**.

Blockchain & Full Stack Developer

July 2021 – Jan 2022

Trillionbits Ltd, Dhaka, Bangladesh

- Developed and integrated blockchain services with REST APIs, ensuring secure and reliable transactions.
- Built full-stack solutions involving backend API development, frontend interaction, and database schema design.
- Worked with smart contracts and blockchain technologies to create decentralized applications.
- Technologies used: Solidity, JavaScript, Node.js, PostgreSQL, MongoDB.

Education

Daffodil International University, Dhaka, Bangladesh, BSc in Computer Science & Engineering

Jan 2017 – Jan 2021

- CGPA: 3.71/4.0
- Ranked #8 among private universities in Bangladesh (HEQEP, UGC)
- Ranked in Top 351-400 (QS World University Ranking)

Publications

- **A Blockchain-Enabled Distributed Advanced Metering Infrastructure Secure Communication (BC-AMI)** July 2022
Nahida Islam, Sazzadur Rahman, Imtiaz Mahmud, *Md. Nur Amin Sifat*, You-Ze Cho
DOI: doi.org/10.3390/app12147274
- **HealthBlock: A Secured Healthcare System Using Blockchain** June 2023
Md. Mahfujur Rahman, *Md. Nur Amin Sifat*, Mostafizur Rahman, Mushfiqur Rahman
DOI: doi.org/10.1007/978-981-19-8032-9_14

Skills

- **Programming Languages:** Python | Go | JavaScript | C | SQL
- **Machine Learning:** LLMs | RAG | Deep Learning | Computer Vision | Agentic AI | Transformer Architecture | CNN | RNN
- **Frameworks:** LangChain | LangGraph | FastAPI | Django | Django Rest API | Go Gin | TensorFlow
- **Databases:** PostgreSQL | MongoDB | Redis | ChromaDB
- **Tools:** Git | Jira | Docker | Docker Compose | Kubernetes
- **Soft Skills:** Effective communication, creative problem-solving, research-oriented thinking

Certifications

- **Deep Learning** — Kaggle
Skills: ANN, SGD, Dropout, Batch, Binary Classification
- **Intermediate Machine Learning** — Kaggle
Skills: Model validation, Feature engineering, Random Forest, XGBoost, Data leakage handling
- **Intro to Machine Learning** — Kaggle
Skills: Supervised learning, Data Preprocessing, Decision trees, Overfitting, Model selection
- **Cyber Security and Physical Protection at Workplace (Malware)** — Brain Station 23
Skills: Malware analysis, Cyber hygiene, Office system protection
- **A Blockchain-Enabled Distributed Advanced Metering Infrastructure Secure Communication (BC-AMI)** — MDPI
Skills: Blockchain, IoT security, Secure communication protocols
- **Python (Intermediate)** — HackerRank
Skills: Data structures, Algorithms, Python functions, OOP, File I/O

Languages

- **English** — Proficient (IELTS Band 6)
- **Bangla** — Native Speaker

Projects

AI Research Agent (Agentic AI & RAG)

- Designed an AI Research Agent to assist researchers in managing and exploring large collections of documents.
- Enabled storage of research documents in a vector database with embedding-based retrieval for precise context matching.

- Implemented a hybrid query pipeline where the system first searches embeddings for relevant content; if none is found, it falls back to direct LLM responses.

Eye Grading Module Web App

- Developed a machine learning-based eye grading module that helps doctors assess eye health based on patient images
- Tools Used: Django Rest API, ML, Reactjs, PostgreSQL

Deep Learning to Predict Human Emotions

- Created a web application for predicting human emotions using deep learning and facial expression recognition
- Tools Used: Python, Django Rest API

Review Management and Suggestions (Agentic AI & RAG)

- Developed a restaurant review management system using Agentic AI and Retrieval-Augmented Generation (RAG) to provide intelligent insights and suggestions.
- Tools Used: LangChain, LangGraph, VectorDB, FastAPI

References

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